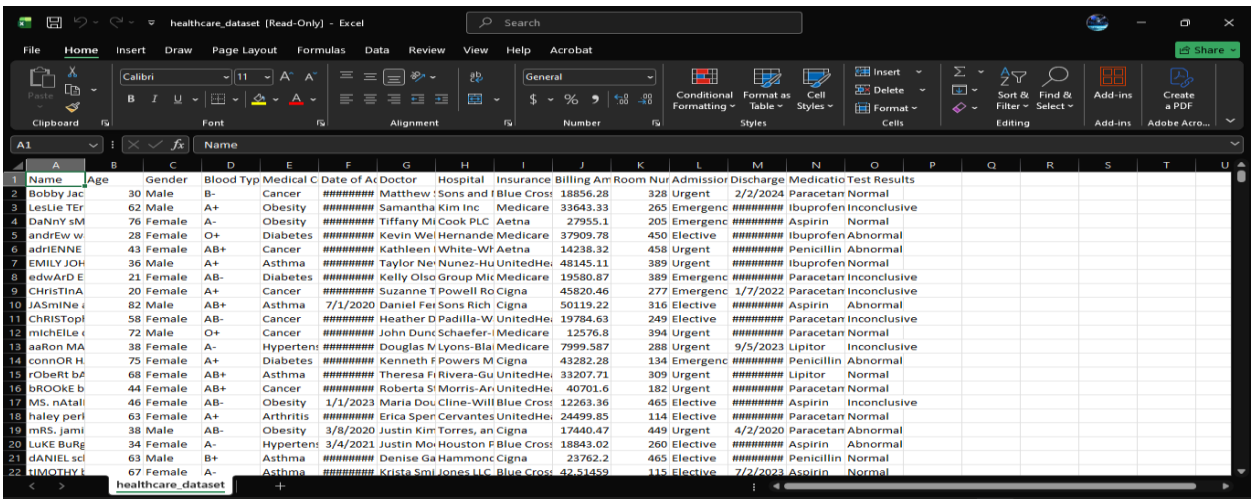
DSA 3050A — Screenshots

PART A — Data Acquisition and Understanding

Raw Dataset in Microsoft Excel

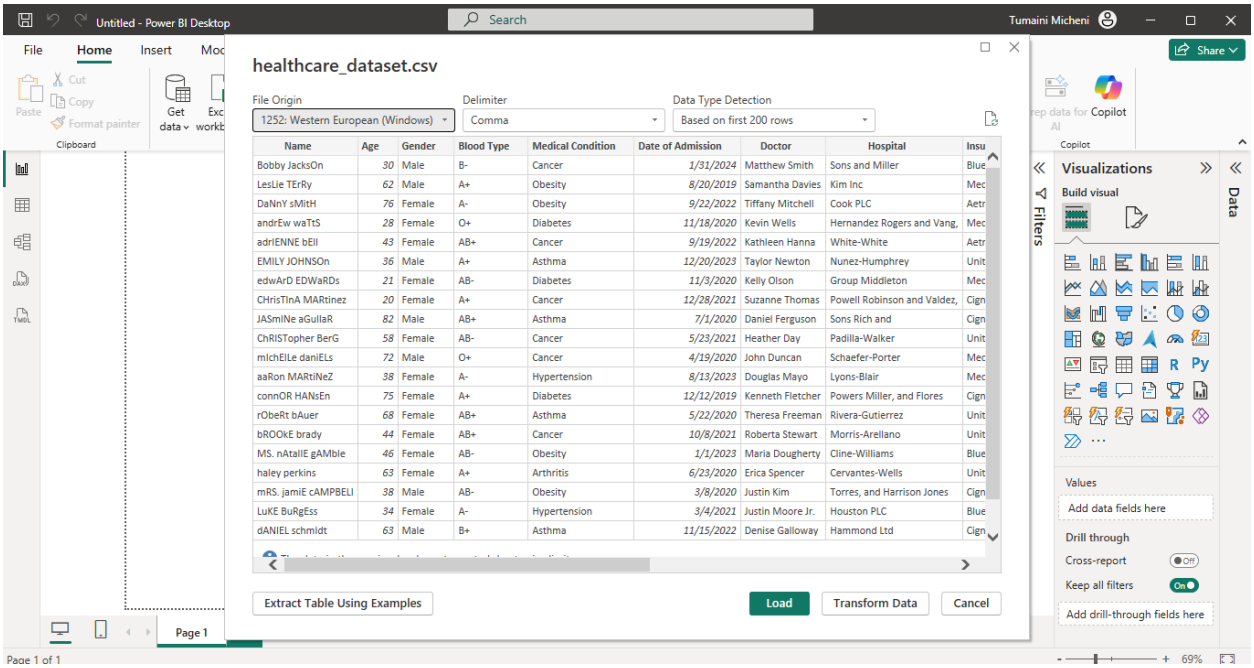
Figure 1: Raw view of the healthcare_dataset.csv file opened in Microsoft Excel, showing all 15 column headers and the first 20 patient records. The dataset contains a mix of text fields (Name, Gender, Medical Condition, Doctor, Hospital, Insurance Provider, Admission Type, Medication, Test Results), numerical fields (Age, Room Number, Billing Amount), and date fields (Date of Admission, Discharge Date). This initial inspection confirms the dataset is well-structured and contains sufficient variables for advanced healthcare analytics. The file contains 55,500+ rows of synthetic patient admission records sourced from Kaggle (prasad22/healthcare-dataset).



Name	Age	Gender	Blood Typ	Medical C	Date of Ac	Doctor	Hospital	Insurance	Billing Am	Room Nur	Admissior	Discharge	Medicatio	Test Results
Bobby Jac	30	Male	B-	Cancer	########	Matthew S	Sons and I	Blue Cross	18856.28	328	Urgent	2/2/2024	Paracetam	Normal
LesLie TEr	62	Male	A+	Obesity	########	Samantha	Kim Inc	Medicare	33643.33	265	Emergenc	########	Ibuprofen	Inconclusive
DaNnY sM	76	Female	A-	Obesity	########	Tiffany Mi	Cook PLC	Aetna	27955.1	205	Emergenc	########	Aspirin	Normal
andrEw w	28	Female	O+	Diabetes	########	Kevin Wel	Hernande	Medicare	37909.78	450	Elective	########	Ibuprofen	Abnormal
adrIENNE	43	Female	AB+	Cancer	########	Kathleen I	White-Wh	Aetna	14238.32	458	Urgent	########	Penicillin	Abnormal
EMILY JOH	36	Male	A+	Asthma	########	Taylor Ne	Nunez-Hu	UnitedHea	48145.11	389	Urgent	########	Ibuprofen	Normal
edwArD E	21	Female	AB-	Diabetes	########	Kelly Olso	Group Mic	Medicare	19580.87	389	Emergenc	########	Paracetam	Inconclusive
CHrisTInA	20	Female	A+	Cancer	########	Suzanne T	Powell Ro	Cigna	45820.46	277	Emergenc	1/7/2022	Paracetam	Inconclusive
JASmINe	82	Male	AB+	Asthma	7/1/2020	Daniel Fer	Sons Rich	Cigna	50119.22	316	Elective	########	Aspirin	Abnormal
ChRISTopl	58	Female	AB-	Cancer	########	Heather D	Padilla-W	UnitedHea	19784.63	249	Elective	########	Paracetam	Inconclusive
mIchElLe	72	Male	O+	Cancer	########	John Dun	Schaefer-I	Medicare	12576.8	394	Urgent	########	Paracetam	Normal
aaRon MA	38	Female	A-	Hyperten	########	Douglas M	Lyons-Bla	Medicare	7999.587	288	Urgent	9/5/2023	Lipitor	Inconclusive
connOR H	75	Female	A+	Diabetes	########	Kenneth F	Powers M	Cigna	43282.28	134	Emergenc	########	Penicillin	Abnormal
rObeRt bA	68	Female	AB+	Asthma	########	Theresa Fi	Rivera-Gu	UnitedHea	33207.71	309	Urgent	########	Lipitor	Normal
bROOkE b	44	Female	AB+	Cancer	########	Roberta S	Morris-Ar	UnitedHea	40701.6	182	Urgent	########	Paracetam	Normal
MS. nAtal	46	Female	AB-	Obesity	1/1/2023	Maria Dou	Cline-Will	Blue Cross	12263.36	465	Elective	########	Aspirin	Inconclusive
haley perl	63	Female	A+	Arthritis	########	Erica Sper	Cervantes	UnitedHea	24499.85	114	Elective	########	Paracetam	Normal
mRS. jami	38	Male	AB-	Obesity	3/8/2020	Justin Kim	Torres, an	Cigna	17440.47	449	Urgent	4/2/2020	Paracetam	Abnormal
LuKE BuRg	34	Female	A-	Hyperten	3/4/2021	Justin Mo	Houston F	Blue Cross	18843.02	260	Elective	########	Aspirin	Abnormal
dANIEL scl	63	Male	B+	Asthma	########	Denise Ga	Hammonc	Cigna	23762.2	465	Elective	########	Penicillin	Normal
tIMOTHY k	67	Female	A-	Asthma	########	Krista Smi	Jones LLC	Blue Cross	42.51459	115	Elective	7/2/2023	Aspirin	Normal

Data Preview in Power BI (Before Transform)

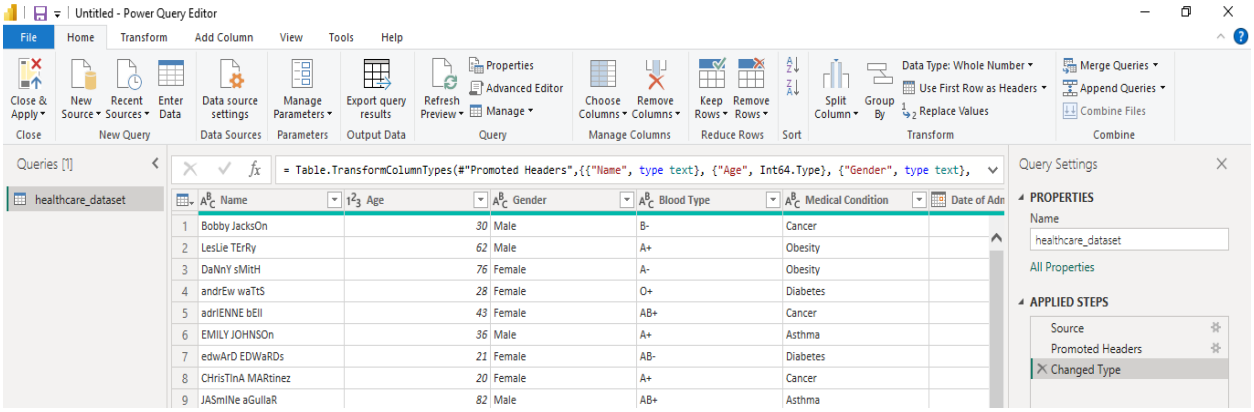
Figure 2: The Power BI data preview window displaying the healthcare_dataset.csv file immediately after import, before any transformations are applied. The preview confirms that Power BI has correctly detected the column delimiter and can read all 15 fields. Key columns visible include Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, and Insurance Provider. The 'Transform Data' button was selected (rather than 'Load') to proceed directly into Power Query Editor for data cleaning and transformation. This step marks the beginning of the ETL (Extract, Transform, Load) process.



PART B — Data Cleaning and Transformation in Power Query

Power Query Editor: Initial Data Load

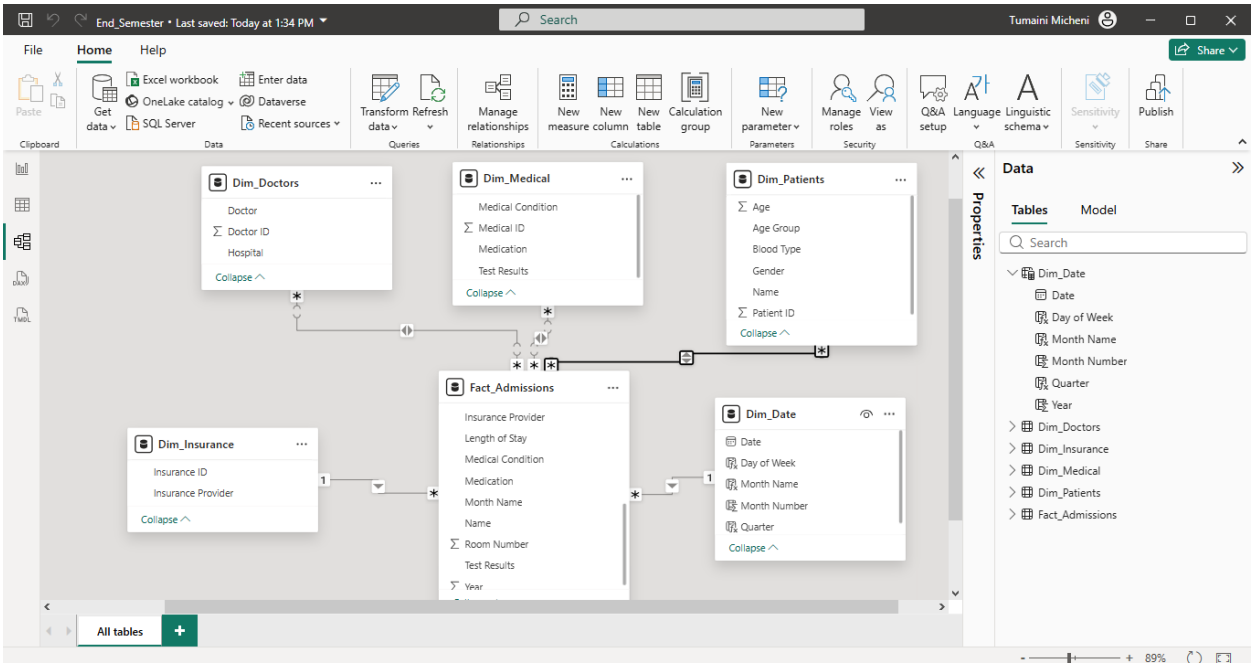
Figure 3: Power Query Editor showing the healthcare_dataset loaded with its initial 'Applied Steps' visible in the Query Settings panel on the right. Three steps have been automatically created: Source (reading the CSV file), Promoted Headers (using the first row as column names), and Changed Type (Power BI's initial automatic data type detection). The column headers are visible at the top of the data grid with their respective data type icons — ABC for text columns and date icons for date columns. This view represents the starting state of the data before any manual cleaning steps are applied. At this stage the dataset has 999+ rows visible in preview with 15 columns.



PART C — Data Modeling

Star Schema Model View

Figure 9: The Model View in Power BI Desktop showing the complete star schema data model after all relationships have been established. The central Fact_Admissions table is connected to four surrounding dimension tables: Dim_Doctors (linked via the Doctor field), Dim_Medical (linked via Medical Condition), Dim_Patients (linked via Name), and Dim_Insurance (linked via Insurance Provider). A separate Dim_Date table is connected to Fact_Admissions via the Admission Date field. All relationships use Many-to-One cardinality — many rows in Fact_Admissions correspond to one row in each dimension table. The Data pane on the right confirms all six tables are loaded and shows the columns within Dim_Date including Date, Day of Week, Month Name, Month Number, Quarter, and Year — confirming the date table has been correctly populated with time intelligence attributes. This model structure enables efficient cross-filtering and time-based analysis throughout the report.

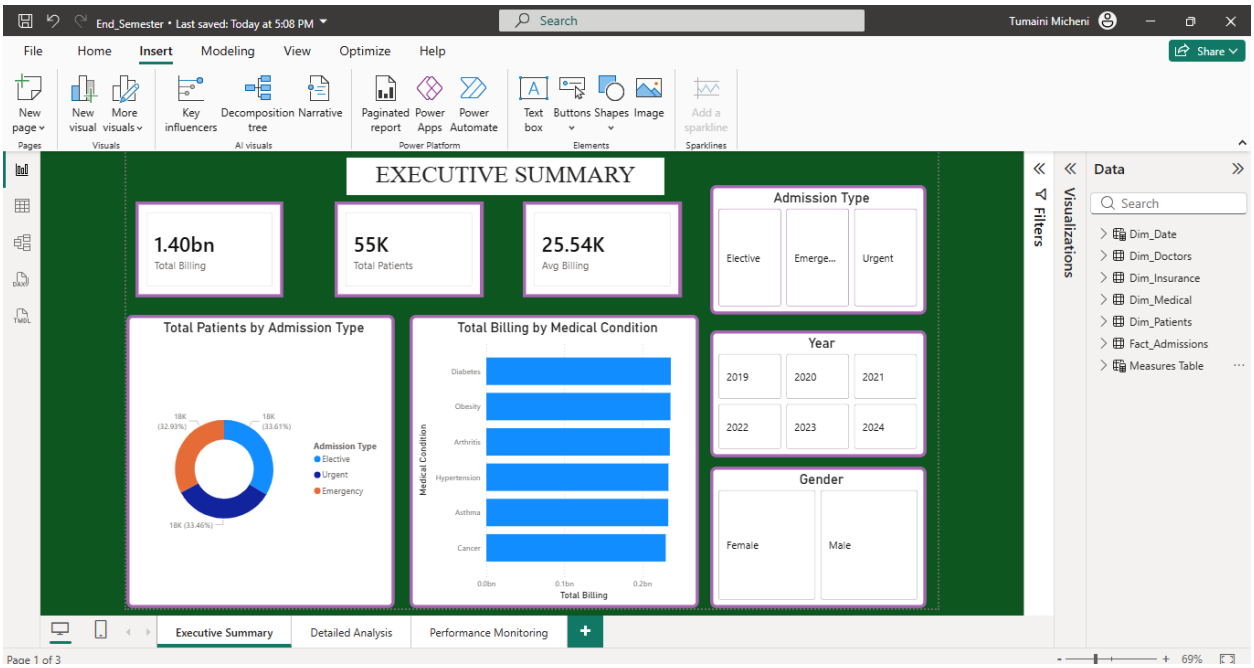


PART E — Dashboard Design and Visualization

Page 1: Executive Summary

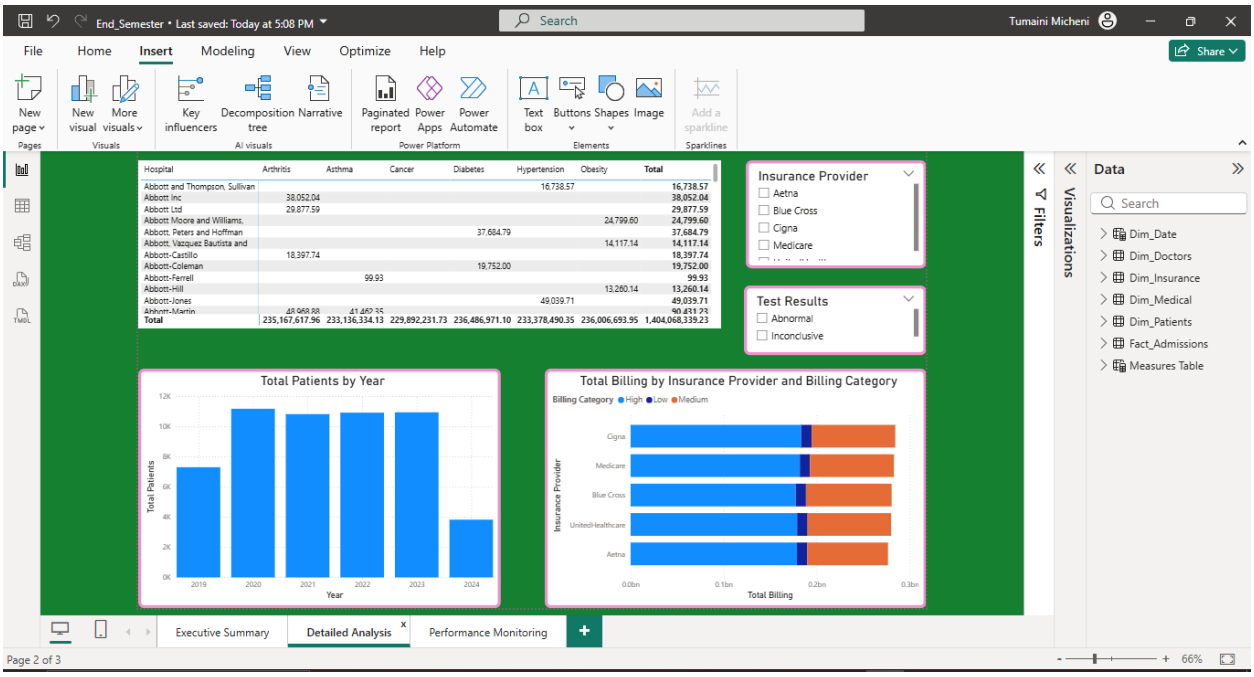
Figure 10: The Executive Summary dashboard page — the first of three report pages. This page provides a high-level overview of key hospital performance metrics. Three KPI cards at the top display the most critical measures: Total Billing of 1.40 billion USD, Total Patients of 55K admissions, and Average Billing per patient of 25.54K USD. A donut chart on the right titled 'Total Patients by Admission Type' shows the near-equal distribution between Elective (32.93%), Emergency (33.46%), and Urgent (33.61%) admissions — revealing no significant imbalance in how patients are being admitted. A horizontal bar chart in the centre titled 'Total Billing by Medical Condition' ranks the six conditions (Diabetes, Obesity, Arthritis, Hypertension, Asthma, Cancer) by total billing, with Diabetes and Obesity generating the highest costs. Two

slicers on the right — 'Admission Type' and 'Year' — allow users to dynamically filter all visuals on the page, and a 'Gender' slicer provides demographic filtering. The dark green background with white and purple visual containers creates a professional, high-contrast executive reporting aesthetic.



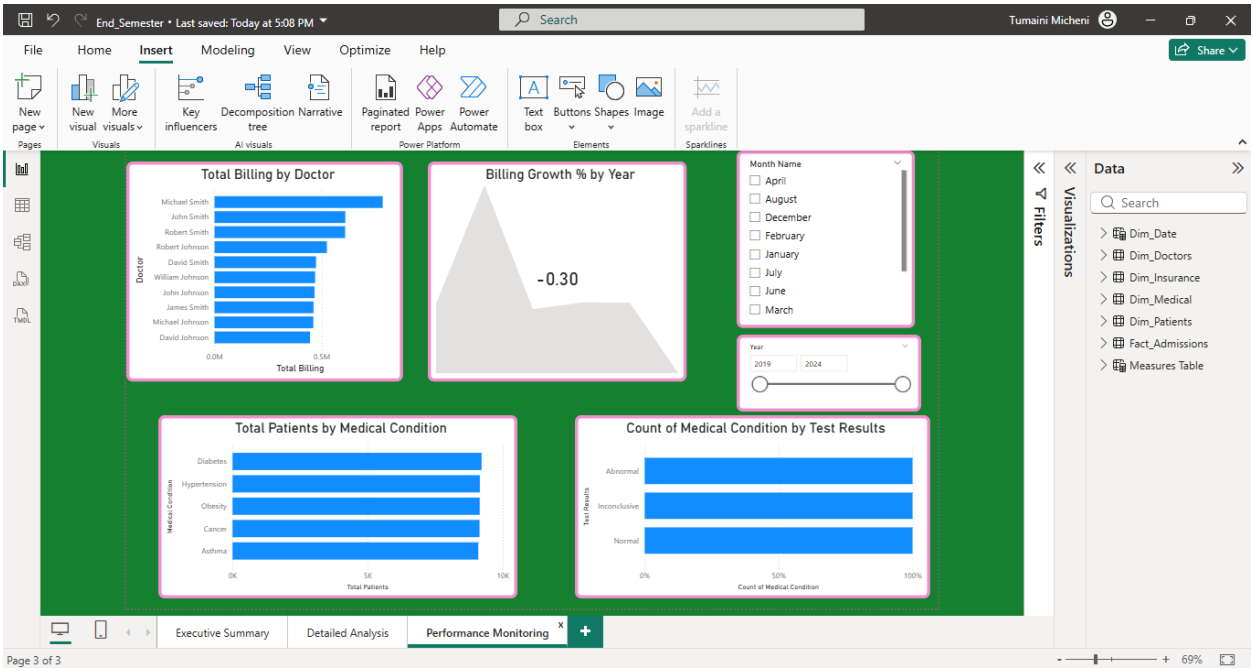
Page 2: Detailed Analysis

Figure 11: The Detailed Analysis dashboard page — the second of three report pages. This page provides a deeper breakdown of billing and patient data across hospitals, years, and insurance providers. The top section features a Matrix Table showing Total Billing by Hospital (rows) and Medical Condition (columns), with grand totals. This allows stakeholders to identify which hospitals are generating the most revenue from specific conditions. Two charts occupy the bottom half: a bar chart titled 'Total Patients by Year' shows patient volume from 2019 to 2024, revealing a significant drop in 2024 (partial year data). A stacked bar chart titled 'Total Billing by Insurance Provider and Billing Category' shows that all five insurance providers (Cigna, Medicare, Blue Cross, UnitedHealthcare, Aetna) generate similar billing totals, predominantly in the High and Medium billing categories, with minimal Low-category billing. Two slicers on the right — 'Insurance Provider' and 'Test Results' — allow users to filter by provider and patient outcome, enabling cross-analysis between financial and clinical data.



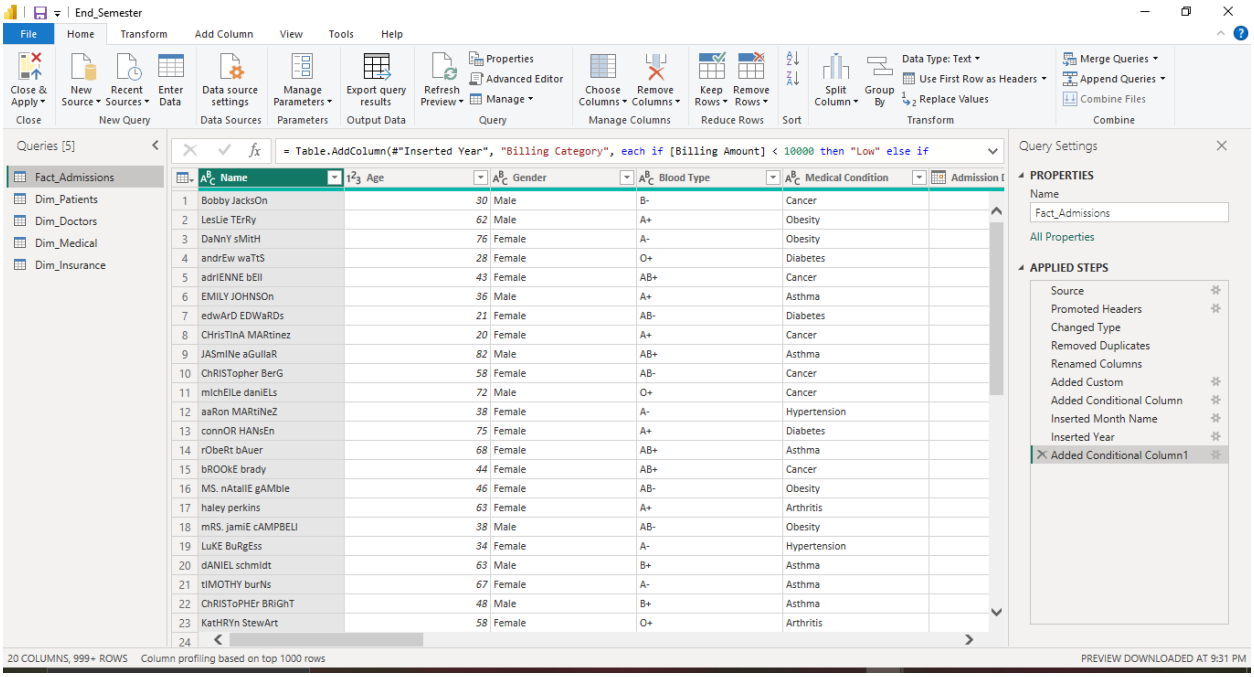
Page 3: Performance Monitoring

Figure 12: The Performance Monitoring dashboard page — the third and final report page. This page focuses on time intelligence, provider performance, and clinical outcome distribution. The top-left horizontal bar chart titled 'Total Billing by Doctor' ranks the top 10 doctors by total billing generated, with Michael Smith, John Smith, and Robert Smith appearing at the top — useful for identifying high-performing or high-activity doctors. An area chart titled 'Billing Growth % by Year' occupies the top centre, currently showing a value of -0.30, indicating a slight year-over-year decline in billing — a key performance indicator that management should monitor. The bottom-left bar chart titled 'Total Patients by Medical Condition' ranks the six medical conditions by patient volume, with Diabetes and Hypertension leading. The bottom-right 100% stacked bar chart titled 'Count of Medical Condition by Test Results' shows the distribution of Abnormal, Inconclusive, and Normal test results across all conditions — revealing an approximately equal split across outcomes for each condition. Two slicers on the right — 'Month Name' and a 'Year' range slider (2019–2024) — enable granular time-based filtering for detailed period analysis.



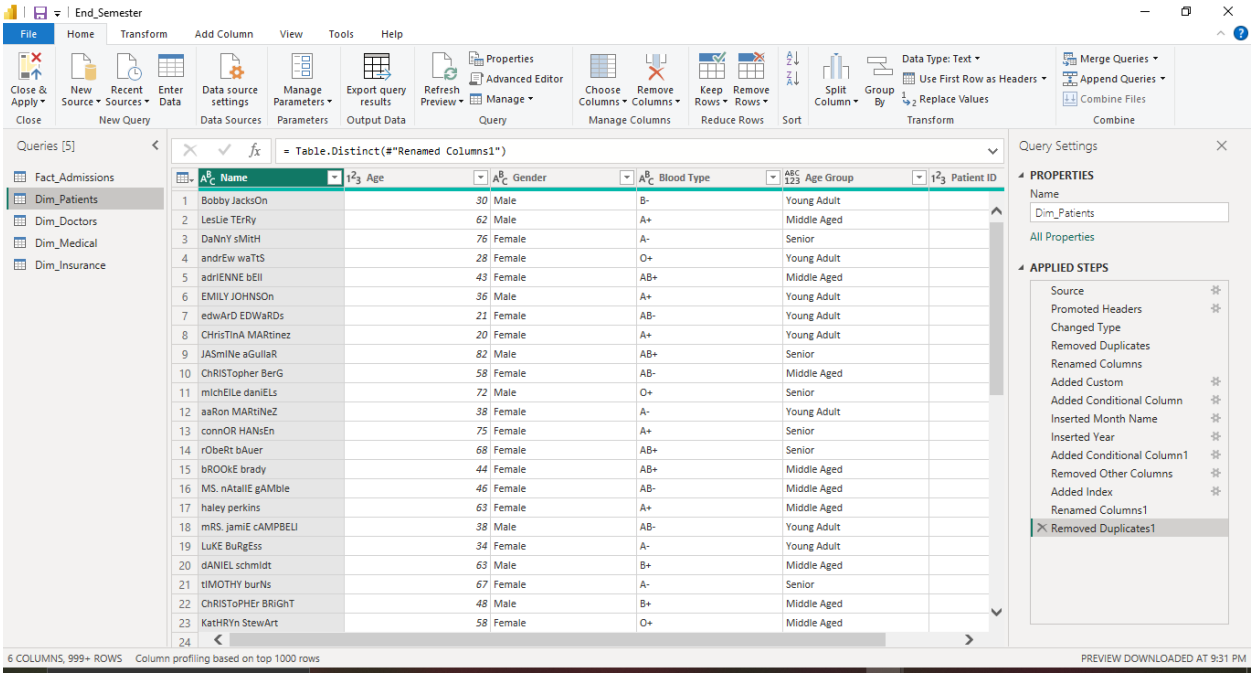
Fact_Admissions Table in Power Query (After All Transformations)

Figure 4: The Fact_Admissions query in Power Query Editor after all transformation steps have been applied. The Applied Steps panel on the right shows the complete transformation history: Source, Promoted Headers, Changed Type, Removed Duplicates, Renamed Columns, Added Custom (Length of Stay), Added Conditional Column (Age Group), Inserted Month Name, Inserted Year, Added Conditional Column 1 (Billing Category). The formula bar at the top shows the M code for the final 'Added Conditional Column1' step which creates the Billing Category column. The table now has 20 columns and 999+ rows visible in preview, with the status bar at the bottom confirming this. The five separate queries (Fact_Admissions, Dim_Patients, Dim_Doctors, Dim_Medical, Dim_Insurance) are visible in the Queries panel on the left, confirming the star schema split has been completed.



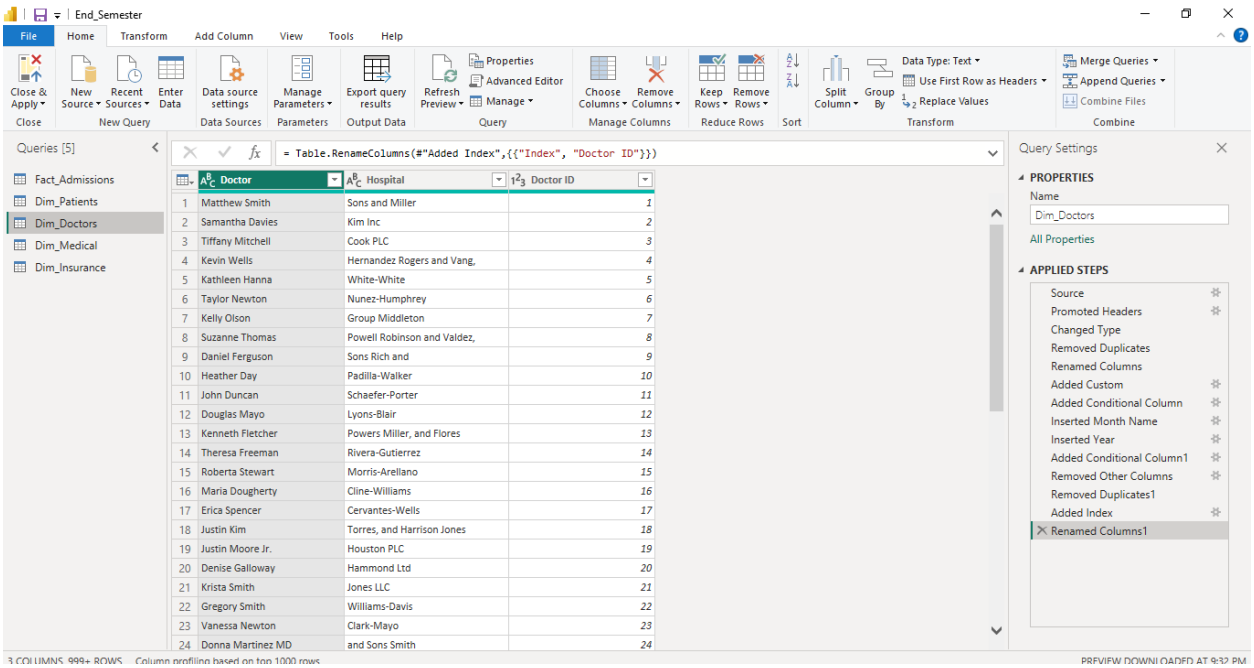
Dim_Patients Table in Power Query

Figure 5: The Dim_Patients dimension table in Power Query Editor, showing the result of duplicating and transforming the original dataset into a focused patient demographics table. The table contains 6 columns: Name, Age, Gender, Blood Type, Age Group (a derived conditional column), and Patient ID (an index column added for unique identification). The Age Group column correctly categorises patients — for example, Bobby JacksOn (age 30) is classified as 'Young Adult', LesLie TErRy (age 62) as 'Middle Aged', and DaNnY sMitH (age 76) as 'Senior'. The Applied Steps panel shows the full transformation pipeline including Removed Duplicates, Renamed Columns, Added Conditional Column, Removed Other Columns, Added Index, and a final Removed Duplicates step to ensure uniqueness. The status bar shows 6 columns and 999+ rows in preview.



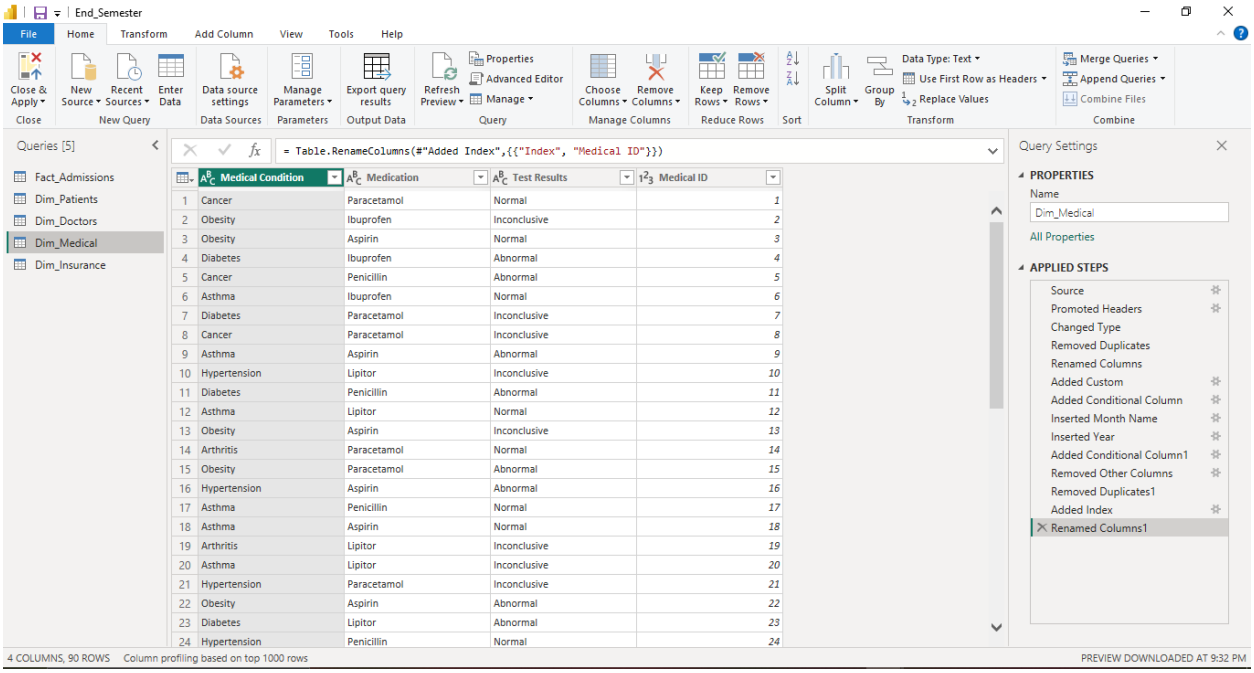
Dim_Doctors Table in Power Query

Figure 6: The Dim_Doctors dimension table in Power Query Editor, containing doctor and hospital pairing information. This table has 3 columns: Doctor (text), Hospital (text), and Doctor ID (whole number index). The table lists individual doctor-hospital pairs — for example, Matthew Smith at Sons and Miller, Samantha Davies at Kim Inc, and Tiffany Mitchell at Cook PLC. The Applied Steps panel on the right mirrors the Fact_Admissions transformation pipeline up to the point where other columns were removed and only the Doctor, Hospital, and index fields were retained. The formula bar shows the final rename step converting 'Index' to 'Doctor ID'. This table serves as the Dim_Doctors dimension in the star schema, enabling analysis of billing and patient volumes by doctor and hospital.



Dim_Medical Table in Power Query

Figure 7: The Dim_Medical dimension table in Power Query Editor, containing medical condition, medication, and test result combinations. This table has 4 columns: Medical Condition, Medication, Test Results, and Medical ID (index). The data shows the variety of condition-medication-result combinations in the dataset — for example, Cancer treated with Paracetamol resulting in Normal test results, Obesity treated with Ibuprofen resulting in Inconclusive results, and Diabetes treated with Ibuprofen resulting in Abnormal results. The status bar shows 4 columns and 90 rows — indicating that after removing duplicates, there are 90 unique Medical Condition, Medication, and Test Result combinations in the dataset. This table enables filtering and cross-analysis of medical conditions in the dashboard.



Dim_Insurance Table in Power Query

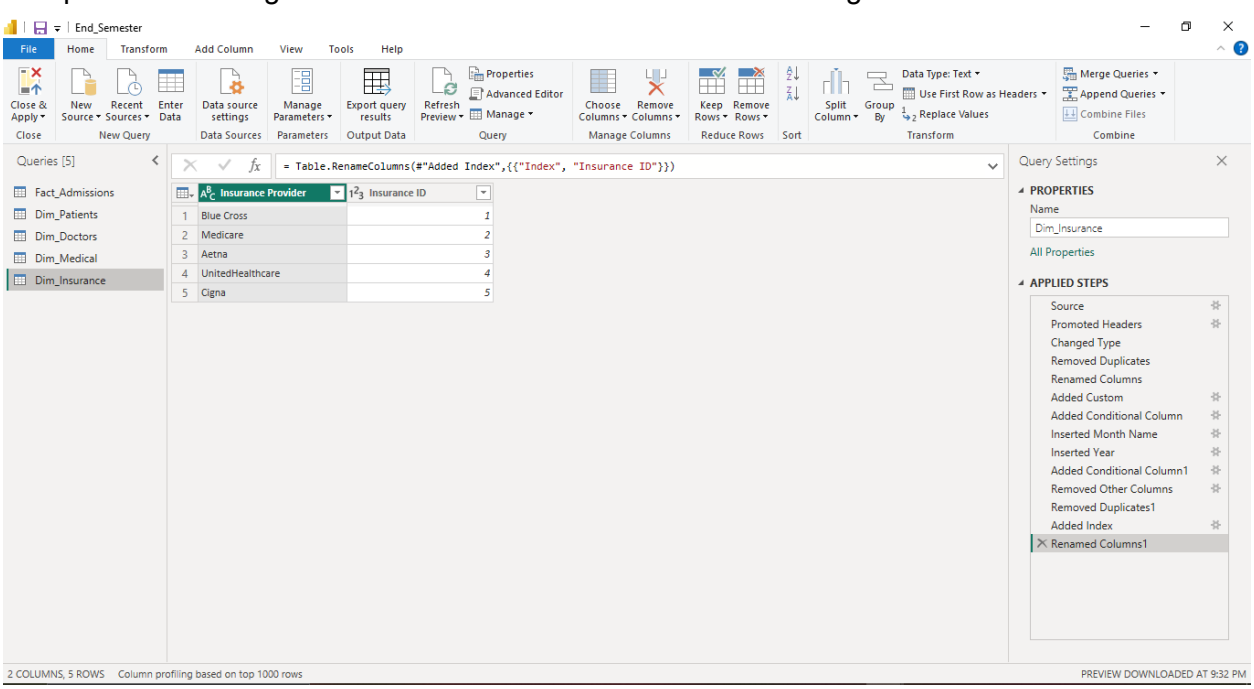



Figure 8: The Dim_Insurance dimension table in Power Query Editor, containing the five unique insurance providers in the dataset alongside their assigned Insurance ID. The table has 2 columns: Insurance Provider (text) and Insurance ID (whole number index). The five providers are: Blue Cross (ID 1), Medicare (ID 2), Aetna (ID 3), UnitedHealthcare (ID 4), and Cigna (ID 5). This is a small but important dimension table — it allows the dashboard to filter all admissions, billing, and patient counts by insurance provider. The simplicity of this table (2 columns, 5 rows) is expected for a single-attribute dimension in a star schema design.